

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: ONE

CLASS: SS3

SUBJECT: Chemistry

TOPIC: Solubility & Salts

Introduction

- Solubility is the maximum amount of a solute that dissolves in a given amount of solvent at a particular temperature, to form a saturated solution. Solubility is often expressed as grams of solute per 100 g (or 100 cm³) of water.

Saturated, unsaturated and supersaturated solutions

- **Saturated** — no more solute will dissolve at that temperature (dynamic equilibrium between dissolving and crystallising).
- **Unsaturated** — more solute can still dissolve.
- **Supersaturated** — contains more dissolved solute than the saturation level; it is unstable and crystallises on seeding.

Solubility curves

- A solubility curve plots solubility against temperature. Most solids become more soluble as temperature rises (the curve rises); gases become less soluble as temperature rises (the curve falls). The curves allow calculation of the mass of crystals that form on cooling.

Preparation of salts

- Salts are prepared by: (1) acid + metal → salt + hydrogen (using a metal above hydrogen in the reactivity series); (2) acid + base/metal oxide → salt + water; (3) acid + alkali → salt + water (titration, for soluble salts); (4) acid + carbonate → salt + water + carbon dioxide; (5) direct combination of elements (e.g. iron + sulphur → iron(II) sulphide). Insoluble salts are made by precipitation (double decomposition). Solubility rules decide the method: all nitrates are soluble; all chlorides are soluble except AgCl, PbCl₂, Hg₂Cl₂; all sulphates are soluble except BaSO₄, PbSO₄, CaSO₄; carbonates, sulphites and hydroxides are insoluble except those of sodium, potassium and ammonium.

CLASS EVALUATION

1. Define solubility.
2. What is a saturated solution?
3. How does temperature affect the solubility of a gas?

4. Which method is used to prepare soluble copper(II) sulphate from insoluble copper(II) oxide?
5. State the solubility rule for chlorides.

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WEEK: TWO

CLASS: SS3

SUBJECT: Chemistry

TOPIC: Volumetric Analysis

Introduction

- Volumetric analysis (titrimetry) is the measurement of the volumes of solutions that react exactly with one another. It is used to find unknown concentrations, e.g. the molarity of an acid or alkali.

Apparatus and procedure

- **Burette** delivers the titrant drop by drop; **pipette** (with a pipette filler) measures a fixed volume of the solution in the conical flask; **conical flask** holds the solution being titrated; **indicator** shows the end point (methyl orange for strong acid–weak alkali; phenolphthalein for strong alkali–weak acid). Read the burette meniscus at eye level; repeat the titration until two concordant titres are obtained (differing by 0.1 cm³ or less); always rinse the burette and pipette with the solutions they will hold.

Calculations

- Key relations: moles = concentration (mol/dm³) x volume (dm³). At the end point, moles of acid ÷ moles of alkali = ratio from the balanced equation; balance the equation first.
Example: $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ (1:1). 25.0 cm³ of NaOH is neutralised by 20.0 cm³ of 0.100 mol/dm³ HCl → moles HCl = 0.100 x 0.020 = 0.002; moles NaOH = 0.002; concentration NaOH = 0.002 ÷ 0.025 = 0.080 mol/dm³.

Standard solutions

- A standard solution has a known concentration, made by dissolving a precisely weighed sample in a known volume of water. The primary standard must be pure, stable and of high molar mass (e.g. anhydrous sodium carbonate for acids); NaOH is not a primary standard because it absorbs moisture and carbon dioxide.

CLASS EVALUATION

1. Define volumetric analysis.
2. State the use of each of: burette, pipette, indicator.
3. What are concordant titres?
4. Why is NaOH not a primary standard?

5. 25.0 cm³ of 0.080 mol/dm³ HCl neutralises 20.0 cm³ of NaOH. Find the molarity of the NaOH.

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WEEK: THREE

CLASS: SS3

SUBJECT: Chemistry

TOPIC: Atomic Theory (Advanced)

Introduction

- The modern atomic theory builds on Dalton's ideas with the discovery of sub-atomic particles and the nucleus. This week deepens the treatment: orbitals, isotopes, electronic structure and the mass spectrometer.

Atomic structure refined

- Atoms consist of protons and neutrons in a very small, dense nucleus, with electrons occupying energy levels (shells) and sub-levels (s, p, d orbitals). The maximum electrons per shell follow $2n^2$ (2, 8, 18, 32). The valence electrons determine chemical properties and the position of the element in the periodic table.

Isotopes

- Isotopes are atoms of the same element with the same proton number but different mass numbers (different numbers of neutrons). They have identical chemical properties but differ in physical properties. Examples: H-1, H-2, H-3; C-12 and C-14; Cl-35 and Cl-37. Isotopes are used in medicine (radioactive I-131 for the thyroid), archaeology (C-14 dating) and industry. Relative atomic mass is the weighted average of the masses of the isotopes — it explains non-integer atomic masses (e.g. Cl = 35.5).

The mass spectrometer

- It separates isotopes by mass/charge ratio: a vapourised sample is ionised by a beam of electrons; the ions are accelerated and deflected by a magnetic field; the lighter ions deflect more, and the detector records a peak for each — the positions give masses, the peak heights give abundances.

CLASS EVALUATION

1. Sketch the electronic structure of chlorine ($Z = 17$) and state its group.
2. Why do isotopes of an element have the same chemical properties?
3. Explain why the atomic mass of chlorine is 35.5.
4. Briefly describe how a mass spectrometer separates isotopes.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FOUR

CLASS: SS3

SUBJECT: Chemistry

TOPIC: Chemical Kinetics

Introduction

- Chemical kinetics is the study of the rates of chemical reactions — how fast reactants turn into products, and which factors control the speed.

Measuring rate

- Rate = change in concentration (or amount) ÷ time. It is measured by tracking the disappearance of a reactant or the appearance of a product over time: volume of gas evolved (by gas syringe), loss in mass (if a gas escapes), colour change, or titration of samples at intervals. The rate decreases with time as reactants are used up — the concentration–time graph is a curve flattening to zero.

Factors affecting rate

- **Concentration** — higher concentration → more frequent collisions → faster (collision theory).
- **Temperature** — a 10 °C rise roughly doubles the rate; particles gain kinetic energy and more collisions have energy above the activation energy.
- **Surface area** — smaller particles (more area) react faster (powdered limestone vs lumps).
- **Catalyst** — lowers activation energy by providing an alternative pathway; the catalyst is unchanged at the end.
- Pressure (for gases) and light (for photochemical reactions) also matter.

Collision theory

- Reactions occur only when particles collide with energy equal to or greater than the activation energy (E_a), and in the correct orientation. Anything that increases the frequency or energy of effective collisions speeds up the reaction.

CLASS EVALUATION

1. Define rate of reaction.
2. State four factors that affect the rate of a reaction with reasons.
3. What is activation energy?

4. Explain why a catalyst speeds up a reaction without being used up.
5. Why do reactions slow down as they proceed?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FIVE

CLASS: SS3

SUBJECT: Chemistry

TOPIC: Equilibrium

Introduction

- Many reactions are reversible: they can go forward and backward. Chemical equilibrium is the state in which the forward and backward rates are equal, so the concentrations of reactants and products remain constant (though both reactions continue).

Dynamic equilibrium

- Equilibrium is dynamic — not static: molecules keep converting in both directions at equal rates. It can be reached only in a closed system. At equilibrium the position (how far towards products) is described by the equilibrium constant, K_{eq} ; $K > 1$ favours products, $K < 1$ favours reactants.

Le Chatelier's principle

- If a system at equilibrium is disturbed (by a change in concentration, pressure or temperature), the equilibrium shifts in the direction that reduces the disturbance.
- Concentration** — adding a reactant shifts right; removing a product shifts right. **Pressure** — increasing pressure shifts towards the side with fewer gas molecules. **Temperature** — raising it favours the endothermic direction; lowering it favours the exothermic direction. **Catalyst** — speeds up the approach to equilibrium but does NOT shift the position.

Application: the Haber process

- $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (exothermic forward). Industrial conditions: 450 °C, 200 atm, iron catalyst. The compromise temperature gives a reasonable rate, the high pressure pushes the equilibrium right, and the catalyst speeds the reaction without changing the yield.

CLASS EVALUATION

- Define dynamic equilibrium.
- State Le Chatelier's principle.
- For $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (exothermic), what happens when pressure is increased and why?
- Does a catalyst increase the yield at equilibrium? Explain.
- Why is 450 °C used in the Haber process if the reaction is exothermic?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: SIX

CLASS: SS3

SUBJECT: Chemistry

TOPIC: Electrolysis & Electrochemical Cells

Introduction

- Electrolysis is the decomposition of a compound (electrolyte) by the passage of an electric current, using inert electrodes. Electrochemical cells, by contrast, convert chemical energy into electrical energy.

Electrolysis principles

- Electrolytes are compounds that conduct electricity in solution or when molten because they contain mobile ions — acids, alkalis and salts. Non-electrolytes (sugar, ethanol) do not conduct. At the cathode, cations gain electrons (reduction); at the anode, anions lose electrons (oxidation); the electrode where oxidation occurs is the anode.

Predicting products

- In molten lead(II) bromide: $\text{Pb}^{2+} + 2\text{e}^{-} \rightarrow \text{Pb}$ (cathode); $2\text{Br}^{-} \rightarrow \text{Br}_2 + 2\text{e}^{-}$ (anode). In dilute aqueous sodium chloride with inert electrodes: H_2 at the cathode and O_2 at the anode because the discharge series selects the ion easier to discharge (in solution, H^{+} before Na^{+} ; OH^{-} before Cl^{-} in dilute solution, but Cl^{-} in concentrated brine). Copper refining uses copper anodes and a copper cathode with copper(II) sulphate electrolyte.

Faraday's laws & uses

- The mass of a substance deposited is proportional to the quantity of electricity passed ($Q = It$), and to the chemical equivalent. Uses: extraction of reactive metals (aluminium, sodium), electroplating (gold/silver plating), refining of copper, production of chlorine and caustic soda, and anodising.

CLASS EVALUATION

1. Define electrolysis.
2. Which ions discharge at each electrode in molten PbBr_2 ?
3. Predict the products of electrolysis of concentrated brine.
4. State four industrial applications of electrolysis.